



MARWADI UNIVERSITY

Faculty of Engineering & Technology

B.TECH ICT

SEM: 5

MID EXAMINATION (OPEN BOOK)

SEPTEMBER: 2025

Subject: - DSIP (01CT1513)

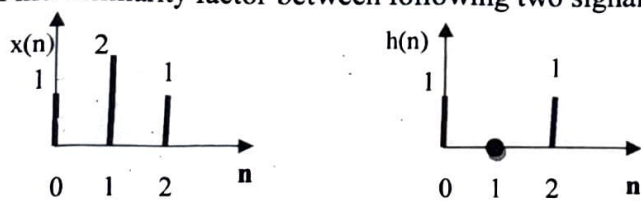
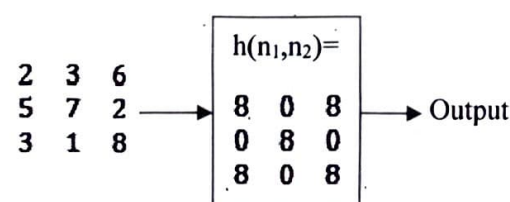
Total Marks:-30

Date:-11-9-2025

Time: - 75 Minutes

Instructions:

1. All Questions are Compulsory.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Do not write/sign/indication/tick mark anything other than Enroll. No. at a specific place on the question paper.

Q. No.	Question	Marks	BL	CO
1	(a) Find the Linear Convolution and Circular Convolution of following signals: $x(n)=3\delta(n)+5\delta(n-1)+2\delta(n-2)$ and $h(n)=6u(n)-3u(n-1)-3u(n-2)$	6	Evaluate	CO3
2	(a) If any LTI System is defined by following input-output relationship $y(n)=2x(n)+3x(n-1)+4x(n-2)+y(n-1)$ Find (a) Impulse response (b) Unit Step Response (c) Frequency response (d) If input is $e^{5n}u(n)$, Find Output. (b) Find Similarity factor between following two signals	8	Apply	CO1, CO4
		4	Apply	CO2
OR				
	(a) If an image $x(n_1, n_2)$ is applied to any Linear system having kernel $h(n_1, n_2)$, Find the processed image output from this system.	8	Apply	CO1, CO4
				
	(b) Apply Median filtering to the image $x(n_1, n_2)=\delta(n_1-2, n_2-1)+3\delta(n_1-3, n_2-1)+5\delta(n_1-1, n_2-2)+7\delta(n_1-2, n_2-3)+8\delta(n_1-3, n_2-3)+5\delta(n_1-2, n_2-2)+2\delta(n_1-1, n_2-3)$ and generate filtered image. Consider kernel or mask size 3×3 .	4	Apply	CO2, CO3
3	(a) Given a grayscale image $f(x, y)$,	8	Analyse	CO2, CO3

	$\begin{matrix} 8 & 6 & 7 \\ 7 & 6 & 7 \\ 8 & 7 & 8 \end{matrix}$			
	Determine: (a) $f(1,2)$ (b) its gray-level resolution, (c) image size, (d) histogram of $f(x,y)$ and (e) the equalized image after histogram equalization and equalized histogram. (b) Compute the Laplacian of the given image and generate the sharpened image by applying Laplacian filtering. $\begin{matrix} 9 & 3 & 6 \\ 5 & 4 & 2 \\ 7 & 1 & 8 \end{matrix}$	4	Apply	CO3,CO4
	OR			
	(a) Explain in detail the differences between the Discrete-Time Fourier Transform (DTFT), Discrete Fourier Transform (DFT), Discrete-Time Fourier Series (DTFS), and Discrete Fourier Series (DFS) in terms of their definitions, periodicity, domains, and applications. After clearly outlining these differences, compute the 8-point Discrete Fourier Transform (DFT) of the discrete-time sequence $x(n)=\{1,2,1,2,1,2,1,2\}$.	8	Analyse	CO2,CO3
	(b) If any signal $x(n)=\{2,0,0,2\}$ and It's DFT $X(k)=\{3.2+j, 1.2-j\}$ Find DFT of $x\{(n-2), \text{Mod } N\}$ and IDFT of $X\{(k-2), \text{Mod } N\}$.	4	Apply	CO2,CO3

---Best of Luck---